

Customer Sales Analysis - Data Analytics Case Study

INTERNSHIP PROJECT 1

Decode Labs Tech

<https://www.decode labs.tech/index.html>

1. Introduction

This project focuses on cleaning and analysing an e-commerce dataset to extract meaningful business insights.

2. Problem Statement

The raw dataset contained missing values, duplicate records, and inconsistent formatting, which required cleaning before analysis.

3. Dataset Description

The dataset contains customer orders including OrderID, Date, CustomerID, Product, Quantity, UnitPrice, PaymentMethod, and TotalPrice.

4. Data Cleaning

(Project 1) Steps performed: - Handled missing values (fillna using mode to only missing categorical values) - Removed duplicate records - Ensured unique Ordered - Converted Date column to proper datetime format - Standardized text fields.

5. Data Analysis / Manipulation / Calculation

(Project 2) Analysis performed: - Sales trend over time - Top selling products - Payment method distribution - Order status insights

6. Visualization Tools used:

- Matplotlib & Seaborn (Python)
- Tableau Dashboard

7. SQL Analysis Used PostgreSQL for querying:

- firstly, I migrate from. ipynb to postgresql and do some basic sql queries to clean csv data files.

8. Key Insights

- Identified high revenue products
- Observed peak sales periods
- Found dominant payment methods

9. Conclusion

The project demonstrates how data cleaning ensures accurate analysis and supports business decision-making.

